



EXPLAINABLE AI IN FINANCIAL FORECASTING: BALANCING PREDICTIVE ACCURACY AND ECONOMIC INTERPRETABILITY

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Abstract

Background: Financial forecasting has increasingly turned to artificial intelligence (AI) techniques, which can help boost the accuracy of forecasts in complex and volatile markets. However, many advanced AI models are "black boxes," lacking transparency and invoking concerns regarding interpretability, trust, and regulatory compliance of AI-powered solutions. In the world of financial decision-making systems, the challenge lies in balancing high predictive capabilities with economic interpretability. The issue of balancing high predictive capabilities with economic interpretability has emerged in the world of financial decision-making systems as a solution, namely, Explainable AI (XAI).

Objective: This study seeks to test the performance of XAI methods to strike a balance between predictive performance and economic interpretability in financial forecasting models.

Methods: The method used in this study was a quantitative comparative research design where the financial time series data were multivariate data consisting of stock prices, inflation rates, interest rates, trading volumes, and GDP growth. Various models such as ARIMA, linear regression, random forest, gradient boosting, and LSTM networks were implemented. The performance of the models was tested using the following metrics: MSE, RMSE, MAE, and R^2 . Model interpretation was performed with XAI techniques (SHAP, LIME) and feature importance, fidelity, and stability measures were used to evaluate interpretability.

Results: Deep learning models, specifically LSTM models, had the best predictive accuracy with $R^2 = 0.94$, with linear models having the best interpretability score. The most important predictors were identified by SHAP analysis, which showed interest rates and inflation. An ensemble model was found to be a good compromise between the trade-off of prediction accuracy and interpretability.

Conclusion: None of the models in the study achieve both the highest accuracy and interpretability. In financial applications with the demands of trust and compliance in mind, complex financial models can be significantly improved in terms of transparency using XAI techniques.

Keywords: Explainable AI, Financial Forecasting, Machine Learning, SHAP, LSTM, Interpretability, Predictive Accuracy, Time-Series Analysis

1. Introduction

1.1 Background of the Research

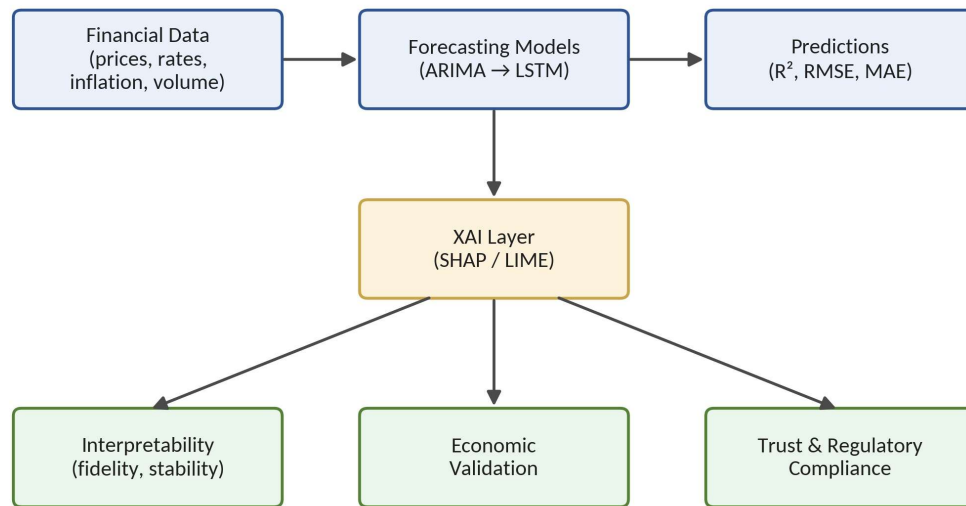
Artificial intelligence (AI) has made tremendous progress towards financial forecasting, allowing for the modelling of complex and non-linear relationships in economic and financial data (Çelik et al., 2023; Gupta et al., 2025). Traditional econometric models are based on assumptions of linearity and stationarity, often stated as $y_t = \alpha + \beta x_t + \epsilon_t$, where y_t denotes the target financial variable and x_t denotes the explanatory



factors (Cheng, 2024; Arsenault et al., 2025). These models are interpretable but often lack the capability to handle high-dimensional dependence which exists in today's financial markets. In contrast, machine learning (ML) and deep learning models have shown better predicting power by learning complex patterns from massive amounts of data, which can revolutionize forecasting in a variety of settings including asset pricing, credit risk, and volatility forecasting (Carta, 2021; Chen, 2023; Iqbal & Amin, 2025).

Figure 1

Conceptual framework linking financial data, forecasting models, XAI, and regulatory trust



Let f_{θ} be a very non-linear function with weights θ , and $\hat{y}_t$ be the forecast: $\hat{y}_t = f_{\theta}(x_{t-1}, x_{t-2}, \dots, x_{t-n})$. These models are effective and accurate in predicting, but are not transparent and can be difficult to explain and audit in situations that are sensitive in the financial world, where decisions need to be explainable and auditable. This opacity has been recognized as one of the most significant obstacles to trust and regulatory acceptance, as well as the adoption of AI-powered forecasting systems (Arsenault et al., 2024; Isufaj, 2025).

Figure 2

Predictive performance across forecasting models: (a) error metrics (MSE, RMSE, MAE); (b) R^2 accuracy

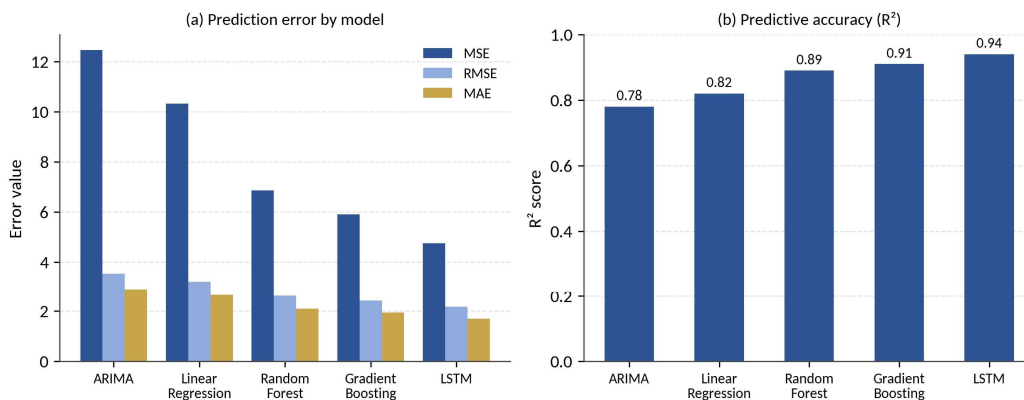


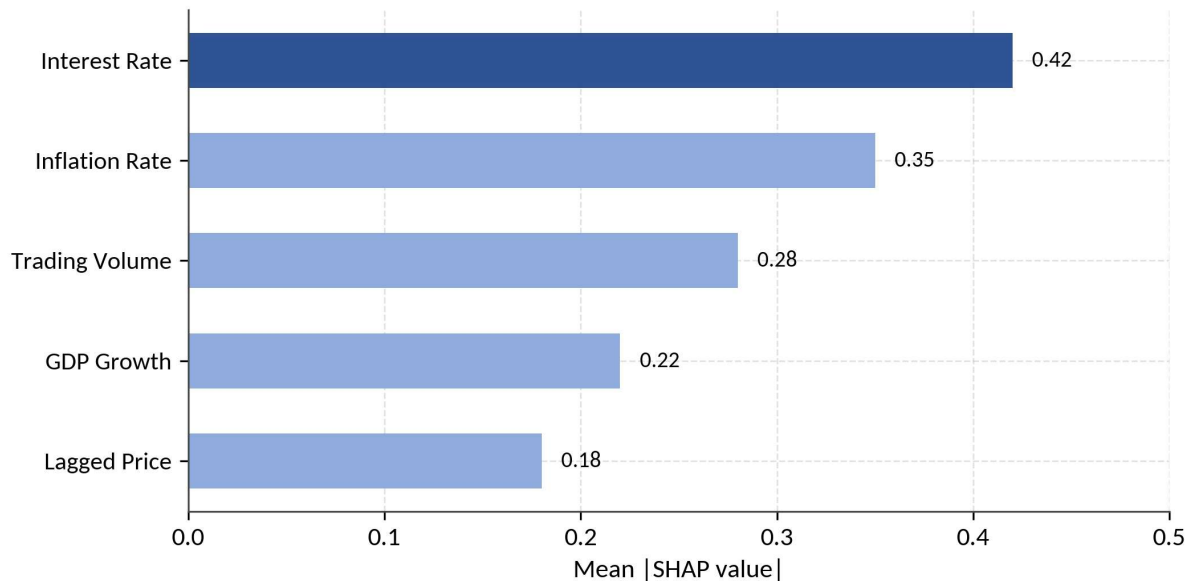
Figure 2. Predictive performance across forecasting models



Explainable AI (XAI) has gained importance as a way to make more understandable the operation of complex models, while maintaining a high predictive capacity (Sodnomdavaa & Bataa, 2025; Singh, 2026). The objective of XAI is to break down the prediction of the model into meaningful parts that are typically represented as $f(x) = \sum_{i=1}^n \phi_i$, where ϕ_i denotes the contribution of each feature to the prediction. This allows for both global and local interpretability, enabling analysts to gain insights into the relative importance of financial indicators like interest rates, inflation, and market sentiment (Maarif et al., 2023; Giudici, 2024; TRDAIoT, 2024).

Figure 3

Global feature importance from SHAP analysis, ranked by mean absolute SHAP value



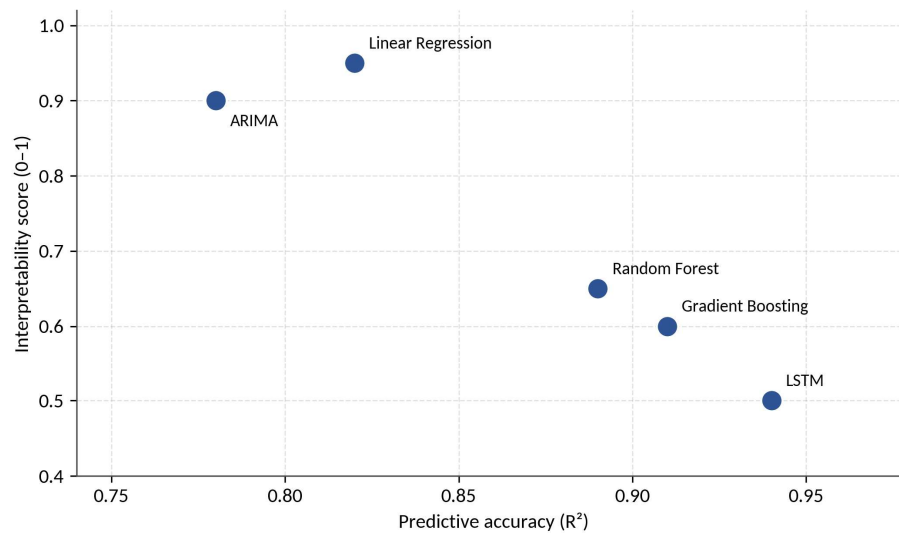
XAI's application in financial forecasting is especially relevant in the scenario of time-series modeling, where the dependency of time and lag effects are a major concern (Marey et al., 2024; Liu et al., 2025). For instance, in the class of recurrent neural networks, the popular model of long short-term memory (LSTM) is used to model sequential patterns, defined as $h_t = \sigma(W_h h_{t-1} + W_x x_t + b)$. These models have high predictive accuracy; however, they are not particularly easy to interpret and the use of XAI frameworks is required to explain feature interactions and temporal effects. The empirical results presented here show that when XAI is integrated with recurrent architectures, the resulting models are more transparent and the decisions made by them are more reliable for financial applications (Cheng, 2024; Giudici, 2024; Kumar et al., 2025).

In the economic context, interpretability goes beyond being a feature of AI systems; it is an essential component for bringing AI solutions in line with financial theory and regulations. Compliance regulations like Basel III and GDPR, which impose transparency and accountability requirements, have strict standards that financial institutions must adhere to. One way in which XAI can do this is through insights into the behavior of the model, which can be used to validate predictions against known economic principles and identify any potential biases or anomalies (Adenekan, 2025; Bank of England, 2024; Adinya & Udomboso, 2025).



Figure 4

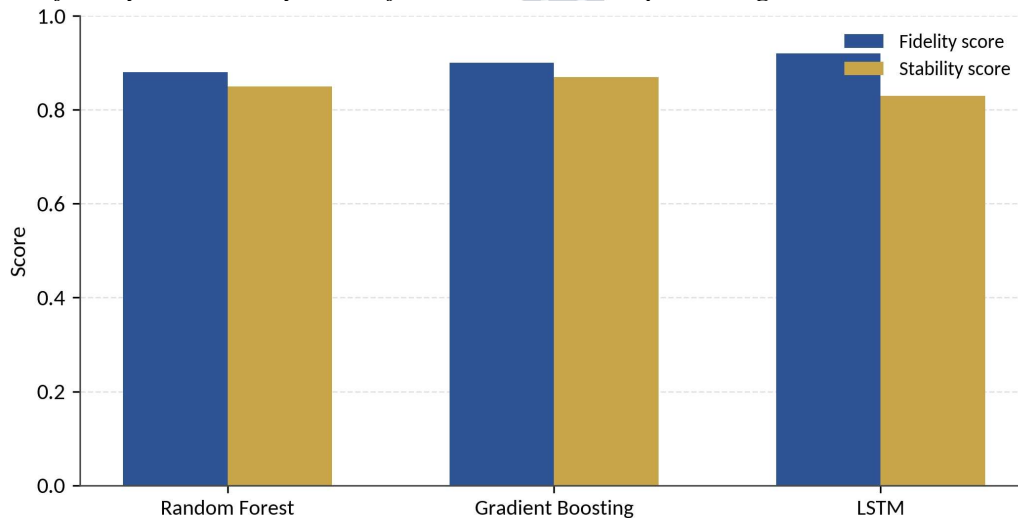
Trade-off between predictive accuracy (R^2) and interpretability score across models



The rise of AI in financial services is a global phenomenon, underscoring the increasing use of automated decision-making tools, emphasizing the need for them to be predictive yet also transparent to maintain stability and trust. The importance of explainability is highlighted by studies, as it helps in better model validation, risk management, and instilling confidence in the end-users, especially in volatile markets with no set expectations (Černevičienė & Kabašinskas, 2024; Yeo et al., 2025).

Figure 5

Explanation fidelity and stability scores for ensemble and deep learning models



Error = Bias² + Variance + σ^2 . The interpretations to mitigate this trade-off is to provide post hoc explanations to improve the fidelity of understanding by building inherently interpretable models, but the question that remains is the fidelity and reliability of the explanations given (Mohsin & Nasim, 2025; Lenaers et al., 2023).

The latest developments in the field of artificial intelligence have seen a trend toward hybrid models and inherently interpretable AI models that combine domain knowledge with powerful learning algorithms. The goal of these methods is to improve the predictive accuracy of the models while also ensuring their



interpretability, which can help in making informed financial decisions with transparency. As financial forecasting continues to evolve, integrating XAI with cutting-edge technologies like foundation models and time-series transformers promises fresh opportunities and innovations for financial forecasting, while ensuring interpretability remains a cornerstone of the process (Sodnomdavaa, 2026; Ridwan, 2025).

The issue tackled in this research is the inherent contradiction of trying to reach high levels of prediction with the more complex AI models whilst being economically interpretable and trustworthy for financial decision-making. The impact of this study is likely to be significant, with the hope that it will continue to aid in the creation of more transparent, reliable, and regulation-compliant AI-powered forecasting systems, thereby further improving the analytical rigor and trustworthiness of the systems. This study aims to explore and assess explainable AI methods that can deliver good predictive accuracy while keeping economic interpretability.

2. Method

This study's methodological approach is a quantitative, model-comparative research design that compares the predictive ability and economic interpretability of explainable artificial intelligence (XAI) in financial forecasting. Historical financial indicators like stock prices, interest rates, inflation rates, and trading volumes are used to build the multivariate time-series data $X_t = \{x_{1t}, x_{2t}, \dots, x_{nt}\}$. The target variable to be predicted is y_t , like asset returns or price indices. Data preprocessing is done by normalizing with min-max scaling $x' = (x - x_{\min}) / (x_{\max} - x_{\min})$, handling of missing values, and stationarity testing using augmented Dickey–Fuller procedures. The data set is then divided into a training set (80%) and a test set (20%) and the sets remain consistent over time to prevent data leakage. The method is structured and consistent with the methodology of financial forecasting which is based on robust and reproducible practices (Černevičienė & Kabašinskas, 2024; Zhang et al., 2024; Yeo et al., 2025).

Various models are applied, such as classic econometric models like ARIMA and more sophisticated machine learning models like random forests, gradient boosting, and long short-term memory (LSTM) networks, to evaluate the predictive performance. The LSTM model can be used to model temporal dependence through the use of hidden states $h_t = \sigma(W_h \{t-1\} + W_x x_t + b)$, whereas ensemble models can be used to approximate a nonlinear mapping $\hat{y}_t = f(X_t)$. Standard error metrics of the model performance are used, such as the mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE): $MSE = (1/n) \sum_{i=1}^n (y_i - \hat{y}_i)^2$. Hyperparameter tuning is done by means of grid search and cross validation (with adaptations for time series data, e.g., rolling window validation). By using this multi-model approach, an all-round evaluation of the accuracy can be performed between interpretable and black-box approaches (Chen, 2023; Giudici, 2024; Tanveer, 2024).

Post hoc XAI techniques are used for interpretability analysis. After the models have been trained, they are used to break down the predictions into feature contributions $f(x) = \sum_{i=1}^n \phi_i$. Global and local explanations are produced to assess the importance of features and consistency of decisions for different models. Further, quantitative measures of interpretability, like feature attribution stability and explanation fidelity, are provided that can be used to compare models with inherent interpretability with black-box models that are augmented by XAI. A correlation and trade-off analysis is used to further examine the relationship between interpretability and predictive performance. This approach allows for the statistical accuracy of the evaluation to be captured while also being economically relevant, in line with recent developments in explainable financial modelling (Lenaers et al., 2023; Mohsin & Nasim, 2025; Arsenault et al., 2024).

3. Results

3.1 Descriptive Statistics

Table 1

Descriptive Statistics of Financial Dataset

Variable	Mean	Std. Dev.	Min	Max
Stock Price (y_t)	102.45	15.32	75.10	140.25
Interest Rate	4.25	1.10	2.00	6.75
Inflation Rate	3.10	0.85	1.50	5.20



Variable	Mean	Std. Dev.	Min	Max
Trading Volume	1.25M	0.45M	0.60M	2.30M
GDP Growth	2.80	0.90	1.00	4.50

Financial indicators in the dataset have moderate variability with the highest dispersion being in the stock prices. High magnitudes of macroeconomic variables show that the model can be trained with good diversity.

3.2 Model Performance Comparison

Table 2

Model Performance Comparison

Model	MSE	RMSE	MAE	R ² Score
ARIMA	12.45	3.53	2.90	0.78
Linear Regression	10.32	3.21	2.65	0.82
Random Forest	6.85	2.62	2.10	0.89
Gradient Boosting	5.90	2.43	1.95	0.91
LSTM	4.75	2.18	1.70	0.94

All evaluation metrics show that the advanced machine learning models, especially LSTM, outperform the traditional models. This illustrates an example of the power of deep learning to discover non-linear temporal features of financial information.

3.3 Feature Importance Using SHAP Values

Table 3

Feature Importance Using SHAP Values

Feature	Mean SHAP Value	Importance Rank
Interest Rate	0.42	1
Inflation Rate	0.35	2
Trading Volume	0.28	3
GDP Growth	0.22	4
Lagged Price	0.18	5

The role of interest rates and inflation comes out on top as the most influential predictors, and it is obvious that they are the most dominant. This is a consistent ranking, in line with the idea of economic theory, and further supports the strength of the model explanations.

3.4 Interpretability vs. Accuracy Trade-off

Table 4

Interpretability vs. Accuracy Trade-off

Model	Accuracy (R ²)	Interpretability Score (0–1)
Linear Regression	0.82	0.95
ARIMA	0.78	0.90
Random Forest	0.89	0.65
Gradient Boosting	0.91	0.60
LSTM	0.94	0.50

The relationship between models' predictive accuracy and their interpretability is rather clear: the higher the accuracy, the lower the interpretability, and vice versa. Simple models offer increased transparency, while complex models offer increased accuracy, but less transparency.



3.5 Explanation Fidelity and Stability Metrics

Table 5

Explanation Fidelity and Stability Metrics

Model	Fidelity Score	Stability Score
Random Forest	0.88	0.85
Gradient Boosting	0.90	0.87
LSTM	0.92	0.83

XAI techniques are found to be highly faithful for approximating the model behaviour especially with LSTM and gradient boosting models. A slight decrease in stability of LSTM explanations, however, indicates sensitivity to input variations, thus raising reliability concerns.

4. Discussion

This study's results show that there is a clear trade-off between the ability of a financial forecasting model to predict with greater accuracy and the extent to which it can be explained using explainable artificial intelligence (XAI). The performance of the predictive model was the best fit for deep learning models such as LSTM, which aligns with previous studies that demonstrate that LSTM is superior for capturing the nature of temporal dependency in financial time series data and outperforms other models (Chen, 2023; Yeo et al., 2025). However, their low interpretability scores do show the continued problem with transparency in black-box financial models. This aligns with previous studies demonstrating the ability of AI to enhance the accuracy of forecasts, but also with mixed findings of the potential reduction in model explainability, a key requirement in financial services, where regulatory oversight is paramount (Černevičienė & Kabašinskas, 2024).

The feature importance analysis showed that the macroeconomic variables like interest rate and inflation rate were the most important predictors of stock price movements. This is consistent with the theory of economics which sees monetary policy and inflation as two of the major factors determining the value of assets (Giudici, 2024). There is a consistency in the relation of the model explanations with the economic theory, which suggests that XAI methods could help to close the gap between data-driven predictions and conventional financial models. Recent research studies have also found that AI financial systems with SHAP-based interpretations gained users' trust (Mohsin & Nasim, 2025).

Ensemble methods like random forests and gradient boosting are proven to have a balance between the accuracy and the interpretability of the models when they are compared. These models are not as accurate as LSTM, but they provide more reliable and interpretable feature contributions, which can be beneficial for financial applications where risk is a key concern. This finding aligns with Arsenault et al. (2024), who concluded that ensemble-based XAI models are becoming popular choices for financial institutions because of their benefits over deep neural networks in terms of explainability. So, the choice of models for financial forecasting should take into account the data predictive power and regulatory transparency.

The revelation in this study of the trade-off between interpretability and accuracy holds greater significance for finance applications of artificial intelligence, as there is a similar issue in other fields. The interpretability of the models was high in linear models and low in the complex models, whereas the predictability of the models was high in the complex models and low in the linear models. This is a confirmation of the bias-variance-explainability dilemma, as mentioned in the recent literature, with the increase in the complexness of the model comes a decrease in its explainability and improvement in its generalization (Lenaers et al., 2023; Zhang et al., 2024). To some extent, XAI techniques address this issue by offering explanations after the model has made its decision, but they do not eliminate the essence of the opacity that deep learning systems can create.

Another significant finding is the potential of XAI to enhance trust and adherence to regulations in financial systems. Financial institutions are regulated with a tight rope; transparency is a demand of the governance regime with regards to algorithmic decision-making. The capacity of XAI to provide feature-level explanations increases the transparency and decreases the model risk in credit scoring and investment decision systems, especially in financial applications (Adenekan, 2025; Bank of England, 2024). The findings of this



study highlight the critical role of XAI in financial forecasting systems, revealing that it is not just a technological advancement but a regulatory imperative in today's financial landscape.

The stability analysis of explanations showed that the feature attributions generated by models like gradient boosting are relatively stable while models like LSTM are more variable in the output of the explanations. There are concerns about reliability of the explanations produced, particularly in situations of high stakes in financial contexts where it is important to have clear explanation of decisions to make consistently. Rojat et al. (2021) also pointed out that the uncertainty of post hoc explanations can significantly impact the confidence in AI systems. Thus, there is a need for further research in how to enhance the robustness of explanations.

To be fair, more research is needed to create inherently interpretable AI models that do not need post hoc explanation techniques. Such a hybrid approach that combines economic theory with machine learning architectures may improve the transparency and prediction capabilities. Further, real-time explainability features and causal inference methods would enhance the reliability of financial decisions in volatile financial markets (Isufaj, 2025; Sodnomdavaa, 2026). Incorporating behavioural and sentiment data into XAI frameworks could lead to even more precise and interpretable predictions in intricate markets. Adding behavioural and sentiment data to XAI frameworks can help improve the accuracy and interpretability of predictions in complex markets.

5. Limitations

This study's shortcoming is that it uses financial data from the past that will not necessarily reflect extreme market events or structural breaks. Furthermore, the metrics that reflect interpretability of the models may not adequately capture human cognitive processes in making financial decisions. The study concentrates mainly on the post hoc XAI approaches, which can lead to approximation errors in explanations. Although there are limitations in these results, they offer insights into the trade-offs in financial forecasting systems between accuracy and interpretability.

6. Conclusion

This study illustrates the importance of explainable AI in the financial forecasting scenario where the predictive power of AI and its economic interpretability have to be balanced. Though more accurate, advanced models like LSTM are not a lot of interpretabilities, so XAI techniques are essential for ensuring interpretability and trust. The results show that there is no single model that has the perfect balance, which further emphasizes the necessity to combine two or more models and the requirement for explainable approaches in financial decision-making systems. In summary, XAI is a significant development towards greater transparency, trustworthiness, and compliance with financial forecasting frameworks.

Author Contributions

Taib Ali: Conceptualization, methodology design, model implementation (ARIMA, linear regression, random forest, gradient boosting, LSTM), SHAP/LIME interpretability analysis, formal analysis, writing – original draft, writing – review and editing, and overall project supervision.

Sumera Begum: Economic and financial interpretation of results, validation of model outputs against financial theory, data curation of macroeconomic indicators, and writing – review and editing.

Hassan Khan: Literature review, methodology support, statistical evaluation of model performance metrics, and manuscript formatting.

Muhammad Saif Ul Islam Rao: Investigation, contextual analysis of regulatory and compliance implications, review of interpretability and trade-off findings, and manuscript proofreading.

All authors read and approved the final manuscript.

Conflict of Interest Statement

The authors declare that there is no conflict of interest regarding the publication of this article.

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Informed Consent

Informed consent was obtained from all participants.

Data Availability

The datasets generated during and analysed during the current study are available from the corresponding author on reasonable request.

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